

With the specification of RNA, the associated DNA can be fetched:

```
>>> my_values_rna = my_values_dna.transcribe()

>>> my_values_rna
Seq('AGUACACUGGU', RNAAlphabet())

>>> my_values_rna.back_transcribe()
Seq('MY_VALUES_DNA', DNAAlphabet())

>>> my_values_rna
Seq('AGUACACUGGU', RNAAlphabet())

>>> my_values_rna.back_transcribe().reverse_complement()
Seq('ACCATGTGACT', DNAAlphabet())
```

Sleep EEG analysis in GNU Octave

Assorted signals are delivered to all parts of the body so that the other organs can communicate with each other for specific or general purposes. One of the key signals in the human brain is electroencephalography (EEG), which is generated from the brain, even when asleep or unconscious. Electroencephalography (EEG) signals comprise brain waves that can be evaluated using GNU Octave. The analysis on sleeping disorders and various diseases can be done with EEG evaluation.

GNU Octave (<https://www.gnu.org/software/octave/>) is one of the powerful and multi-functional tools used for engineering and scientific applications of research. The simulations related to engineering as well as medicine can be implemented with the assorted tool boxes and functions in Octave. It is used as an effective alternate to MATLAB since it is open source and can be freely distributed. A number of tool boxes for different applications are available in GNU Octave, which can be used for optimisation and predictive analysis.

The **Wave Form Database (WFDB)** package can be integrated with GNU Octave. This package is equipped with the functions and modules for EEG and brain signal evaluations. A similar process is followed in case of brain mapping or brain fingerprinting for criminal investigation when the subject is in an unconscious state. There are assorted stages of sleep or unconscious states which can be analysed from EEG signals after recording from the electrodes. This process assists in the forensic analysis of the person while in the unconscious state. By this evaluation, the medical disorders can also be detected using the WFDB package in Octave. The following are the excerpts of Benchmark Sleep Stages which can be evaluated using the WFDB package in GNU Octave so that the overall state of the nervous system can be evaluated and predictions made, along with diagnosing brain disorders.

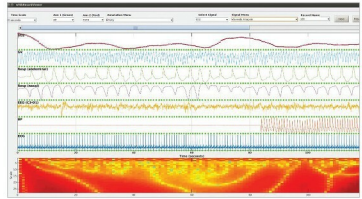


Figure 3: Viewing EEG signals in the WFDB tool box

Stage 1: Tiredness, drowsiness, the pre-sleep stage and lethargy

- Eye activities
- Rolling eye movements
- Sharp transients

Stage 2: Normal night sleep

- Sleep spindles
- Slow eye movement

Stage 3: Delta sleep or slow wave sleep

- Sleep time of 6.5 hours

With the following instruction, the demonstration of the WFDB tool box can be viewed in Octave.

```
>>> wfdbdemo
```

The following instructions can be executed to read and plot the ECG signal from the data set repository of PhysioBank:

```
[time,signal]=rdsamp('mitdb/100',1);
plot(time,signal);
```

(Source: <https://www.physionet.org/physiotools/matlab/wfdb-app-matlab/>)

Using similar methodology, the waveform of arterial blood pressure (ABP) can be analysed using the *wabp* function.

Scope of research in biomedical engineering

Nowadays, bioinformatics and biomedical predictive analytics are two key domains of research for assorted applications. The extraction, processing and predictive mining from the brain, heart and other human body generated signals are evaluated with the use of information technology. The data sets from Physionet, UCSD, FPMS and others can be used for the research work in bioinformatics with the integration of data mining and machine learning tools. **END**

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